Yen-Nan (Aaron) Ho

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AI Engineer

Data Scientist at Harbor Tech, building industrial AI products from signal analysis and model validation through software integration and customer delivery. Lead product/domain design, an auditable AI-assisted issue-repair workflow, and an end-to-end equipment-inspection project, working with AI agents and engineering partners.

Core Skills

- **Industrial AI/Research:** remaining useful life (RUL), phase-resolved partial discharge (PRPD)/vibration, FFT, anomaly detection, CNNs, ablations, confidence, and data-availability assessment.
- **AI Systems:** LangChain/LangGraph, report services, vLLM, OpenAI-compatible APIs, deterministic facts, fallbacks, and human validation.
- **Platform/Delivery:** Python, Django/Django Ninja, PostgreSQL, Redis, RabbitMQ, REST, async workflows, React/TypeScript, Docker, offline deployment, update/rollback, Git, and automated tests.

Work Experience

Data Scientist, Harbor Tech | Taipei | Oct 2025-Present

- **Product/domain architecture:** Led Harbor AI 2.0 product logic, translating partial discharge, RUL, trends, and data quality into equipment status and user flows. Traced a statistic to prevent sampling-point density being presented as discharge-event rate; corrected health calculations that included disconnected equipment. Engineering partners jointly integrated the Platform Core.
- **Model assumptions and validation:** Audited AI-generated RUL conclusions. Traced the data source, found test constants misrepresented as observations, and invalidated the conclusion. Then required multi-parameter, multi-curve, and forward/reverse ablations to recheck the assumptions. Connected research notebooks to production code to reduce drift. Separated fault type, severity, RUL, and low-confidence states.
- **Report-service integration:** Turned field-analysis and manually reviewed reporting methods into reusable workflows. Used a LangGraph POC to separate analysis results from LLM narrative, then advanced the platform-connected report service; DOCX reports and frontend downloads were integrated into the mainline. Separately validated live vLLM endpoint capabilities.
- **Platform and delivery:** Collaborated on Django async workflows, defining APIs, repository/artifact-by-reference contracts, task lifecycles, and validation boundaries. Integrated Docker/offline and cross-architecture installation, platform update/rollback, and auditable migration; led release integration and cut decisions.
- **AI-assisted issue repair and project tracking:** Built a workflow linking daily automated tests and monitoring to issue records, isolated repair branches, AI-assisted implementation and validation, and Bitbucket PRs for human review. Connected issues to verification, decisions, and release records for auditable project tracking. I define specifications, validation criteria, and domain decisions; reviewers and I decide merge and release.

- **End-of-line inspection:** Led my first end-to-end customer project, from requirements and specifications through AI-assisted implementation, UI/UX refinement, and on-site delivery. Worked with engineering partners to adapt long-term monitoring to successive-unit inspection. Demonstrated the system, trained users, and collected feedback on October 12, 2026; formal acceptance remains unconfirmed.

Data Science Intern, Institute of Sociology, Academia Sinica | Taipei | May 2019-Dec 2020

Used survey-area grouping, point-to-point distances, OR-Tools TSP, and Folium for field-survey routes, dispersion, and budget estimation.

Selected Research and Projects

- **MCIENet, M.S. thesis (2024):** Independently led research decomposition, multi-scale CNNs, data pipelines, ablations, cross-dataset evaluation, and explainable AI (XAI) for DNA-sequence-based 3D chromatin interaction prediction. Adapted parts of ChINN preprocessing; retained cross-cell-line generalization limits.
- **Horus, undergraduate capstone (2020):** Team-integrated multi-camera tracking, person ReID, and face recognition. Primarily owned ReID integration, similarity/dual-threshold decisions, and tracking-record workflows. First-place undergraduate capstone.
- **Taipei Disaster Resource Dashboard (2025):** Proposed the topic and collected data for a team project on disaster-prevention resources around tourist attractions. Finalist, Taipei-New Taipei City Dashboard Hackathon; the public dashboard is based on a fork of a friend' s template.
- **PropGMAN (2022):** A three-person team applied spatiotemporal modeling to regional land indicators from property transactions, retaining experiments and reproduction steps. Creative Award, AI for Asset Activation Innovation Contest.

Research and Publications

- Dynamic traffic network representation model for improving the prediction performance of passenger flow. Knowledge-Based Systems, 2024.
- A Dynamic Embedding Method for Passenger Flow Estimation. IIAI AAI, 2021; Competitive Paper Award.
- Hierarchical Attention Network with Sentiment Lexicons for Stock Article Emotion Analysis. TCSE, 2020.

Education

- National Chengchi University, M.S. in Computer Science | Sep 2021-Sep 2024.
- Soochow University, B.S. in Data Science | Sep 2017-Jul 2021.